

## Chain of thought prompt

Scenario: Customer tries to transfer ₹50,000. Transaction failed. Error shown: "Transaction could not be completed."

Act as a Senior Banking Support Investigator

[Mandatory] Let's think step by step to identify the root cause of the failure.

## Result

As a Senior Banking Support Investigator, I would perform a structured investigation by validating each layer involved in the transaction. The generic error message, "**Transaction could not be completed,**" does not identify the actual cause, so the investigation should systematically eliminate possible failures.

### Step 1: Gather Transaction Details

Collect the following information before starting the investigation:

- Transaction ID (if generated)
- Date and Time of transaction
- Customer ID/Account Number
- Source Account
- Beneficiary Account
- Transfer Type (IMPS/NEFT/RTGS/UPI/Internal Transfer)
- Transfer Amount (₹50,000)
- Channel used (Mobile App, Internet Banking, Branch, API)
- Error code (if available in backend logs)
- Device/Browser/App version
- Customer's location/IP address

### Expected Outcome:

- Confirm whether the transaction request actually reached the banking system.

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### Step 2: Verify Customer Account

Check:

- Is the account active?
- Is the account frozen?

- Is there a debit block?
- Is KYC valid?
- Is the account closed?

#### Possible Root Cause

| Validation      | Result                 |
|-----------------|------------------------|
| Account Frozen  | Transaction rejected   |
| Debit Freeze    | Debit not permitted    |
| KYC Expired     | Regulatory restriction |
| Dormant Account | Debit blocked          |

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#### Step 3: Verify Available Balance

Check

Available Balance

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Lien Amount

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Minimum Balance Requirement

Example

Available Balance

₹52,000

Minimum Balance

₹5,000

Transfer

₹50,000

Remaining

₹2,000

Since minimum balance is violated,

**Root Cause**

Insufficient withdrawable balance.

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#### **Step 4: Verify Transaction Limits**

Check configured limits.

Examples:

- Daily Transfer Limit
- Per Transaction Limit
- Monthly Limit
- Beneficiary Limit

Example

Customer Daily Limit

₹25,000

Attempted

₹50,000

Result

Transaction rejected.

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#### **Step 5: Verify Beneficiary**

Validate

- Beneficiary exists
- Account number correct
- IFSC valid
- Beneficiary activated
- Cooling period completed

Possible issue

Beneficiary added recently.

Cooling period

24 Hours

Transfer attempted immediately.

System blocks transaction.

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### **Step 6: Verify Fraud/Risk Engine**

Risk engine may block the transaction.

Check

- Velocity rules
- Unusual location
- Device mismatch
- High-value transfer
- AML rules
- Sanctions screening

Possible scenario

Customer usually transfers

₹5,000

Suddenly

₹50,000

Fraud engine flags transaction.

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### **Step 7: Verify Authentication**

Check

- OTP validated
- Transaction PIN
- MPIN
- Biometric
- Session timeout

Possible failures

- OTP expired
- Wrong OTP
- Token expired

- Session timeout
- 

### **Step 8: Verify Payment Switch**

Check whether request reached

- IMPS Switch
- NEFT Gateway
- RTGS Gateway
- UPI Switch

Possible issues

- Switch unavailable
  - Network timeout
  - Queue overflow
  - Host unavailable
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### **Step 9: Verify Core Banking System (CBS)**

Check CBS logs.

Possible failures

- Debit posting failure
  - Database timeout
  - Lock timeout
  - Ledger unavailable
  - CBS maintenance
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### **Step 10: Verify Database**

Check

- Deadlocks
- Connection pool exhaustion
- Transaction rollback
- Commit failure

Example

Debit Successful

Credit Failed

Rollback Executed

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### **Step 11: Verify Network Infrastructure**

Investigate

- API Gateway
- Firewall
- Load Balancer
- DNS
- VPN
- Internal Network

Possible issues

- Packet loss
  - High latency
  - Gateway timeout
- 

### **Step 12: Verify Application Logs**

Review logs for:

- Exception stack traces
- HTTP status codes
- Internal error codes
- Correlation IDs
- Timeout events

Example

HTTP 504

Gateway Timeout

or

CBS\_CONNECTION\_TIMEOUT

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### Step 13: Verify Third-Party Dependencies

If applicable, verify

- NPCI connectivity
  - Payment gateway availability
  - SMS gateway (OTP delivery)
  - Authentication service
  - Notification service
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### Step 14: Verify Reconciliation Status

Determine whether

- Debit happened?
- Credit happened?
- Transaction pending?
- Auto reversal initiated?

Possible scenarios

| Debit | Credit | Status |
|-------|--------|--------|
|-------|--------|--------|

|    |    |        |
|----|----|--------|
| No | No | Failed |
|----|----|--------|

|     |     |         |
|-----|-----|---------|
| Yes | Yes | Success |
|-----|-----|---------|

|     |    |                   |
|-----|----|-------------------|
| Yes | No | Reversal Required |
|-----|----|-------------------|

|     |         |                      |
|-----|---------|----------------------|
| Yes | Pending | Await Reconciliation |
|-----|---------|----------------------|

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### Step 15: Review Monitoring Alerts

Check dashboards for

- Server CPU

- Memory utilization
- Database health
- Queue backlog
- API latency
- Payment switch alerts

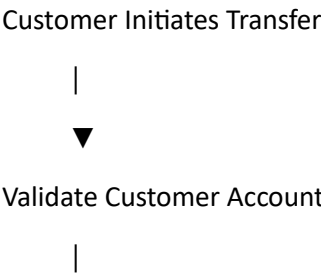
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**Likely Root Causes**

| Category       | Example Root Cause                |
|----------------|-----------------------------------|
| Customer       | Insufficient available balance    |
| Customer       | Daily transfer limit exceeded     |
| Customer       | Beneficiary cooling period        |
| Customer       | Invalid OTP                       |
| Customer       | KYC restriction                   |
| Fraud          | Risk engine blocked transaction   |
| Network        | Gateway timeout                   |
| Database       | Deadlock or rollback              |
| CBS            | Core banking unavailable          |
| Payment Switch | NPCI/NEFT/IMPS switch unavailable |
| Infrastructure | Server overload                   |
| External       | Third-party outage                |

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**Investigation Flow Summary**







Check Available Balance



Check Transaction Limits



Validate Beneficiary



Verify Authentication



Fraud & Risk Assessment



Payment Switch Connectivity



Core Banking Processing



Database Commit



Reconciliation



Determine Root Cause

**Recommended Next Actions**

1. Check whether a transaction ID and backend error code were generated.
2. Confirm if the source account was debited or if the transaction failed before debit.
3. Review application, payment switch, and CBS logs using the transaction timestamp and correlation ID.
4. Validate account balance, transfer limits, beneficiary status, and authentication logs.
5. If the debit occurred without a corresponding credit, prioritize reconciliation and initiate an auto-reversal or manual recovery according to the bank's operational procedures.